

Wynton Zhang

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EDUCATION

Duke University | Durham, NC — Expected May 2028

Bachelor of Science in Engineering, Electrical and Computer Engineering

Minor: Machine Learning and Artificial Intelligence

Relevant coursework: Completed: Data Structures and Algorithms; Signals and Systems; Microelectronic Devices and Circuits. In progress: Robotics and Automation; Digital Systems; Elements of Machine Learning.

TECHNICAL PROJECTS

Robot-Arm Planning and Manipulation | Python, ROS 2, MoveIt, Gazebo, RViz — Fall 2026

- Extended course scaffolds to implement Cartesian placement and multi-block stacking for a simulated Kinova Gen3 Lite arm, coordinating gripper release and planning-scene attachment/detachment.
- Compared baseline and obstacle-avoiding trajectories; added measured gripper-position checks and a recovery sequence that opens, retreats, and stops after a rejected grasp.

Sensor-Based Mobile Robot | Arduino Mega, PID, RFID | Coursework

- Programmed sensor-based PID control to steer a robot along a straight black line and regulate motion; integrated an RFID reader to retrieve and display tag data.

Atrump Mill Oil-Chip Stopper | Onshape, CAD, Technical Drawing — Fall 2026 · In progress

- Created CAD and technical-drawing documentation for a removable mill oil-chip stopper, incorporating a ramped profile and underside reliefs to clear raised machine-bed features.

EXPERIENCE

Software Engineering Intern | Duke | Durham, NC — Jun. 2025 – Aug. 2025

- Implemented a Variational Quantum Eigensolver in Qiskit and benchmarked results against Hartree–Fock and exact diagonalization.
- Compared classical optimizers and presented hybrid quantum–classical workflows to colleagues and Duke Quantum Information Society members.

Software Engineering Intern | Bluestamp | Aurora, CO — Jul. 2023 – Aug. 2024

- Developed Arduino and Raspberry Pi projects and an OpenCV computer-vision pipeline for robotics demonstrations; presented designs to an engineering cohort.

Product Management Intern | Greenhouse Scholars | Boulder, CO — Aug. 2024 – Present

- Developed data pipelines for scholar reporting and coordinated cross-functional teams to deliver student-engagement product features.

Electrical Engineering Intern | Eaglecrest High School | Arapahoe County, CO — Aug. 2022 – Apr. 2023

- Integrated LIDAR and ultrasonic sensors for object detection, created Matplotlib visualizations, and fabricated custom components for competition teams.

TECHNICAL SKILLS

Languages: Python, Java, Verilog, Bash

Robotics & Embedded: ROS 2, MoveIt, Gazebo, RViz, Arduino Mega/Uno, Raspberry Pi, PID control

Tools & Libraries: Git, Linux, Docker, OpenCV, NumPy, Pandas, Matplotlib, Qiskit

Design & Manufacturing: Onshape, Fusion, technical drawings, 3D printing, CNC milling